Elsevier Editorial System(tm) for Agriculture, Ecosystems and Environment
Manuscript Draft

Manuscript Number: AGEE12570R1

Title: Biochar increases vineyard productivity without affecting grape quality: results from a four years field experiment in Tuscany

Article Type: Research Paper

Keywords: biochar; climate change adaptation; sustainable viticulture; grape quality.

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Manuscript Region of Origin: ITALY

Abstract: Biochar application to agricultural soils has proved to substantially modify the plant-soil-water relationship and lead mostly to a quantitative increase in agricultural production through physical, chemical and biological mechanisms. Nevertheless, the impact of biochar on qualitative traits of agricultural production needs to be further assessed.The effect of biochar application on vine yield and grape quality parameters is here investigated in a non-irrigated vineyard in Tuscany (central Italy). Results from four harvest-years showed a higher productivity, up to 66%, of treated plots with respect to their controls, while no significant differences were observed in grape quality parameters. The observed increase in productivity was inversely correlated with rainfall in the vegetative period, confirming the key role of biochar in regulating plant water availability. These findings support the feasibility of a biochar-based strategy as an effective adaptation measure to reduce the impact of water stress periods with no negative effects on grape quality.

Response to review

Ref.: Ms. No. AGEE12570
Biochar increases vineyard productivity without affecting grape quality: results from a four years field experiment in Tuscany
Research Paper
Agriculture, Ecosystems and Environment

Editors' comments:

I have now received two sets of comments on your manuscript (AGEE12570) and these differ widely in their recommendation. Reviewer 1 was very positive but listed some specific points that need your attention please. Reviewer 2 while suggesting the research was topical and potentially interesting also raised concern over the statistical approach that had been adopted.
I have also read your manuscript and have made various suggestions/comments on an attached pdf (some of these have already been raised by reviewer 1).

> *All your comments in the pdf file have been taken into account in the revised version; some of them are addressed in detail below.*

It is perhaps the way you refer to a comparison of biochar 'rates' which made reviewer 2 consider a regression analysis would be more appropriate. The authors should think about removing reference to 'rate', really you only have two treatments with different quantities of biochar applied which in the end showed few actual difference and your ANOVA would seem an appropriate analysis. It would be helpful if you could consider this point please and modify the text accordingly and then place the application rate (20 or 22 t/ha (the amount differs in the text) into some sort of context. Why was this rate chosen, would 10 t/ha have been adequate? Then list some specific objectives at the end of the Introduction.

> *We are aware that the term "rate" could have been misleading. Our aim was not to define a dose-response curve of biochar, but the scope of the second biochar application (indeed a re-application one year later on part of the experimental layout) was to rule out eventual biochar transient effects that could disappear after the first year as observed by other authors (i.e. Quillian et al, 2012).*
> *To clarify this aspect we added a sentence at the end on the introduction and in the discussion section and replaced the word "rate" with "treatment" through the text where opportune. Furthermore, to highlight the "re-application" objective of the second treatment, now we refer to it as 22+22 t ha^{-1} instead of 44 t ha^{-1}.*
> *The biochar rate was chosen arbitrarily within the range of rates recommended by the International Biochar Initiative (5-50 t/ha). Our experiment did not enable to define if a lower dose could have been adequate, but now we highlighted in the discussion that the saturation threshold of biochar benefits is likely to be positioned in our conditions well below 22t ha^{-1}.*

Is it possible to mention some of the costs (see comment from reviewer 1) involved and how the required rate might be site and soil dependent? Perhaps if the effect is mainly related to physical properties then perhaps a single application might be sufficient to provide a long-term benefit but also the size range of the material added might be important.

> *We feel that adding an analysis of costs is a bit out of scope for our work, furthermore, biochar is still not allowed in Italian agriculture (except for research purposes) and, as a consequence, there is still large uncertainty on availability and on the marketable price.*
> *Of course biochar application effect is site specific and biochar type dependent and this concept has been further stressed in the discussion section.*

Concerning the need to highlight the observed long-term benefits, we agree with the editor and a the concept was added both in the result (141-143) and in the discussion (216-219).

The highlights need to be a maximum of 85 characters (including spaces)

Ok, done

Line. 101 is this 500°C?

Yes, we corrected

Response to comment on line 30:

For "qualitative traits" we mean those parameters that define the quality of agricultural production. If unclear, we can eventually modify in "qualitative aspects"

Response to comment on line 36:

Biochar was proven to have an impact on both soil water content and plant available water (Novak et al., 2012; Bruun et al., 2014; Baronti et al., 2014)

Response to comment on line 79: might also be worth mentioning that while irrigation is a potential management option there are various potential issues of this with respect to slope erosion and impact on groundwaster quality.

Yes we agree, we added a couple of words on that, the issue is tackled again in the discussion section

Response to comments at lines 134-135:

Yes, we agree. The sentence was rephrased removing the reference to "rates" and referring to as "re-application treatment". The suggestion on the long-lasting effects was also taken into account in the new version

Response to comment at line 142:

The sentence was rephrased

Response to comment at line 145:

We agree, the part at lines 147-147 was moved to the discussion section

Response to comment at line 168:

We agree, the sentence was rephrased

Response to comment at line 171:

We agree, the sentence was rephrased at lines 178-180 to be in better connection with the following part on the importance of water stress timing on tissues formation.

Response to comment at line 187:

No, we did not expect to have differences in soil temperature between the two treatments as the effect of biochar on surface albedo tends to saturate; in an experiment on wheat we observed no significant differences in the albedo of bare soil between 30 and 60 t ha^{-1}

(Genesio et al, 2012).

Response to comment at Line 228: do you also need to mention that the effects appeared to last 4 years at least?

Yes, the sentence was rephrased and expanded. All this section has been moved upward in the discussion at lines 213-219.

Response to comment on Line 249:

The increased grape fertility refers to the number of seeds per berry that was significantly higher in treated plots. The sentence was rephrased for clarity.

Reviewer #1:
This four-year field experiment is exemplary. The somewhat surprising results are promising for the application of higher amount of biochar in perene cultures in summer dry regions. The paper is excellent and the reviewer has only some minor corrections and propositions:

L28: there are many biochar trials that did not lead to quantitative yield increases, please don't write "generally". You may write: …and lead mostly to…

Ok done

L38 write rather: no negative effects on grape quality of production.

Ok done

L55: omit "on trees"; omit the comma after "and"

Ok done

L56 add "only" in between "particular" and "few"

Ok done

L59 add "systems" after "different soils/climate"

Ok done

L68 add "in" between "vintage" and "respect"

Ok, we added “with” instead of “in”

L73 write "winegrowers" instead of "viticulturist"

Ok, done

L75 write rather "who applied" instead of applying (as you don't do it anymore)

Ok, done

L76-77 write: reported an increase in soil water content, a reduction … and an increase…

Ok, done

L80 end the sentence after production. Start a new sentence with "The only attempt…."

Ok, done

L82 write: (2014) who applied low biochar rates … near Valais and did not observe … nor grape quality…

Ok, done

L94 please give the amount of biochar in DM

Water content of biochar at the time of application was 25% therefore the Dry Matter of biochar applied was 16.5 and 33 t/ha; we added this information in M&M.

L100 please describe more detailed the mode of culture in this vineyard: distance of plantation, canopy height, guyot ?, if green cover what green cover and what percentage of green cover, use of herbicide (how much, how often, when), soil labor (how often, when). Is the soil compacted?

The description of the vineyard and of the management practices was revised at lines 91-99. The description of the fertilization rates refers to a typical management in the study period.

L107 end sentence after "density"

Ok, done.

L116 "individually" instead of singularly

Ok, done.

L152-154 I would propose to move the sentence on seed number before the paragraph on grape quality.

We would prefer to keep the sequence as it is because the grape quality analysis refers to the four harvests and is therefore consistent with the grape production data listed in the first part of results, while the analysis of seeds refers to only two years of the experiment

L196 as you write about the effect on the control rather than of the variants, the sentence is a bit difficult. To make it more clear please add: the berries of the control vines did not show… concentration compared to the berries of the biochar treated plots.

Yes we agree, we corrected and now is more clear, thank you

L209: proven not proved

Ok, done

L211: imbalance not unbalance

Ok, done

L212: polyphenols accumulation,

Ok, done

L212: optimal harvest time.

Ok, done

L213: For this reason This is why irrigation…

Ok, done

L214 through not trough

Ok, done

L214 delete "the" after "through"

Ok, done

L216 move "water" after "civil and industrial"

Ok, done

L220 add "and" after "strategy" instead of comma

Ok, done

L222 write "without" instead of "with not "

Ok, done

L226 I don't agree with the conclusion to use better more than less even if not necessary. The price for the application of 40 t / ha is about 30.000 Eur (including transport and application) which is close to the price of the land in most regions and exceeds the price of annual harvest by about factor three. It is not clear if the saturation effect is close to 22t/ha, it might even be much lower than that. The amount of biochar to reach the saturation effect in a given soil should be subject of further research. The lowest possible amount should proposed to the winegrowers. It would be good to give a short economical estimation of the proposed application method and amount.

We agree with the reviewer that our study does not enable the identification of a saturation threshold. Moreover, these results are very specific to the conditions of our experiment and we agree that it is not appropriate to provide suggestions that could be generalized to other situations. As a consequence this part of the discussion was revised at lines 213-219.

Concerning the economic analysis, we feel that it goes beyond the scope of our work, furthermore the use of biochar is still not allowed in Italy (except for research purposes), and as a consequence large uncertainties remain on its potential availability and consequently on its marketable price. In Italy biochar is currently produced by some pyrogassification plants, but is considered as a sub-product of the energy production process and is treated as a waste; pyrolysis plants spend up to 200€/ton to move this waste to landfills. This does not happen in other countries, in example in the UK where Shakley and coautors (2011) estimated the cost for biochar applied to the field to range from 142 to 289 £ per ton for small pyrolysis plants and this cost tend to be lower and also negative in case of large pyrolysis plants with particular feedstock.

L231 Move this first sentence to the discussion or delete it.

We prefer to maintain this sentence here as a short introduction to the discussion. Instead, as suggested, we shortened the conclusions moving the last part to the discussion section.

L246 - 255: seems to fit better in the discussion than in the conclusion which should be shorter

Yes we agree, all this part was moved at the end of the discussion section

L248 in different areas regions and soils.

Ok, done

L250 other "than" not "that"

Ok, done

Reviewer 2# The article presents a very interesting and important topic.
However, some significant mistakes were done in the statistical analisys.
The article presents results of an experiment with different rates of biochar applied to soil. Despite this, the statistical analysis was done using means comparison tests.
When someone are testing rates of something (biochar, fertilizer, water etc.) the correct form to analyse the data is using regression analysis.
So we are not recommending this article to publication in AGEE.
We suggest the authors change the statistical analysis and then re-submit the article to AGEE.

Please see our reply to the Editor's comments

References

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Shackley, S., Hammond, J., Gaunt, J., & Ibarrola, R. (2011). The feasibility and costs of biochar deployment in the UK. Carbon Management, 2(3), 335-356.

Highlights:

- Biochar was applied to a non irrigated vineyard in Central Italy
- Grape yield increased significantly across four harvests in biochar treated plots
- Yield increase was inversely correlated with rainfall
- No significant difference was found in grape quality parameters
- Biochar can be a suitable strategy for vineyard adaptation to water shortages

*Manuscript
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Biochar increases vineyard productivity without affecting grape quality: results from a four years field experiment in Tuscany

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Abstract

Biochar application to agricultural soils has proved to substantially modify the plant-soil-water relationship and lead mostly to a quantitative increase in agricultural production through physical, chemical and biological mechanisms. Nevertheless, the impact of biochar on qualitative traits of agricultural production needs to be further assessed.

The effect of biochar application on vine yield and grape quality parameters is here investigated in a non-irrigated vineyard in Tuscany (central Italy). Results from four harvest-years showed a higher productivity, up to 66%, of treated plots with respect to their controls, while no significant differences were observed in grape quality parameters. The observed increase in productivity was inversely correlated with rainfall in the vegetative period, confirming the key role of biochar in regulating plant water availability. These findings support the feasibility of a biochar-based strategy as an effective adaptation measure to reduce the impact of water stress periods with no negative effects on grape quality.

Keywords: biochar; climate change adaptation; sustainable viticulture; grape quality.

1. Introduction

Biochar application to agricultural soils has recently emerged as an effective win-win strategy to steadily sequester carbon into agricultural soils, produce renewable energy and increase crop yields (Woolf et al., 2010). The mechanisms involved in the agronomic benefits of biochar are chemical, physical and biological. Biochar is known to decrease nutrients leaching (Lehmann et al., 2013) and diminish the bioavailability of heavy metals (Park et al., 2011), improve soil water holding capacity (Glaser et al., 2002) and plant water availability (Baronti et al., 2014), improve soil structure (Case et al., 2012) and stimulate soil microbial activity (Kolb et al., 2009; Rutigliano et al., 2014), in general leading to increases in crop productivity that have been estimated to be as high as 10% across different crops, soils, biochar types and application rates (Jeffery et al., 2011). It is worth noting that despite many experiments observed positive effects of biochar application on crops, most of them do not refer to long-term experiments (Lorenz and Lal, 2014) and in some cases biochar benefits demonstrated to have a transient nature (Quilliam et al. 2012).

Furthermore, although a quantity of literature has been produced describing the agronomic effect of biochar on herbaceous crops, few studies were dedicated to tree crops, because of the difficulties to perform representative experiments in controlled environment, and the longer time required to produce detectable effects on species with a largely developed root system. In particular only few studies exist on the impact of biochar on vineyards in the primary literature (Baronti et al., 2014; Schmidt et al., 2014), while the importance of enlarging our knowledge on the response of different crops in different soils/climate systems to the application of different biochars has been continuously stressed and identified as a research priority (Mukherjee & Lal, 2014).

Moreover, few researches have assessed the impact on quality of production. Vaccari and co-authors (2011) in an experiment on durum wheat in Central Italy observed an increase up to 30% in above ground biomass and yield without any significant effect on grain quality defined as grain protein content.

68 The aspect of quality has a particular relevance for viticulture that is known to be highly sensitive to
69 inter-annual climate variability. Grape quality parameters substantially affect the value of
70 production, delineate the distinctive character of a *terroir* and define the rating of a vintage with
71 respect to another. Moreover in recent decades global climate changes have been a major cause of
72 concern between winegrowers because of the rise in mean air temperature and of the increase of
73 frequency and intensity of extreme weather events such as droughts and heat-waves. Those changes
74 are projected to increase in the forthcoming decades causing a shift in the viticulture suitability of
75 many productive regions (Hannah et al., 2013; White et al., 2006) and forcing winegrowers to adopt
76 reactive and/or anticipatory adaptation measures (Nicholas and Durham, 2012).
77 Baronti and coauthors (2014), who applied a large volume of biochar for two consecutive seasons
78 to a non irrigated vineyard in Tuscany (central Italy), reported an increase in soil water content, a
79 reduction of plant water stress and an increase of photosynthetic activity during drought. This
80 suggests that the application of biochar to vineyards is a feasible adaptation strategy to reduce the
81 impact of severe water stress periods without recurring to irrigation. Nevertheless, despite these
82 positive results some concerns still remain on the potential impact of such a strategy on quality of
83 production. The only attempt to explore the relation between biochar application and grape quality
84 was made by Schmidt and coauthors (2014) who applied low biochar rates (8 t ha^{-1}) to a vineyard
85 near Valais and did not observe significant variations in plant growth nor changes in grape quality
86 parameters.
87 The present work aims to assess the impact of biochar on vine yield and grape quality parameters in
88 four harvests and is based on the same field experiment described in Baronti et al. (2014). In order
89 to rule out eventual transient effects, after a first application, biochar was re-applied on part of the
90 experimental layout one year later.
91

2. Materials and Methods

The experiment was carried out on a vineyard at "La Braccesca Estate" (Marchesi Antinori srl, www.antinori.it) near Montepulciano in Central Italy (Lat. 43◦10′15″N; Long. 11◦57′ 43″E; elevation 290 m above sea level). The vineyard (*Vitis vinifera* [L]) was planted in 1995 (cv. Merlot, clone 181; rootstock 3309 Couderc) and the trellis system is a single curtain with plant-row spacing of 0.8 m and 2.5 m, respectively. Rows orientation is East–West, inter-rows are partially covered with spontaneous grass, and tilled with a chisel plow in the March-June period. The vineyard is not irrigated and it is fertilized with an inorganic fertilizer (15.0.26) twice per year (in November and April) at a rate of 120 kg ha^{-1}. Soil is acid, shallow and sandy-clay-loam textured (USDA, 2005) (Table 1) and is highly compacted below 0.4 m depth.

Biochar was applied with two treatments, in five replicates randomly distributed, as follows: 22 t ha^{-1} of biochar applied in 2009 (B); 22 t ha^{-1} in 2009 and further 22 t ha^{-1} in 2010 (BB) and control untreated plots (C). Water content of biochar was 25%, therefore each application corresponded to 16.5 t ha^{-1} of dry biochar. Each plot (15 in total) had a surface area of 225 m^2 (7.5 m in width and 30 m in length) including 4 rows and 3 inter-rows. Biochar was superficially applied to soil of vineyard inter-row with a spreader and partially buried with a chisel plow tiller to 0.3 m depth.

Meteorological parameters were registered by an automatic weather station placed nearby the vineyard (Figure 1).

The biochar used in the experiment is a commercial low temperature (500°) slow pyrolisis biochar derived from orchard pruning feedstock provided by Romagna Carbone (Bagnacavallo, Ravenna, Italy). Biochar chemical and physical characteristics are summarized in Table 2, analytical methodologies for biochar characterization are provided in Baronti et al. (2014).

A first experiment has been made on this vineyard specifically focused on soil-plant-water relationships (Baronti et al., 2014). For these purposes soil was sampled 4 times in 2011 to assess gravimetrically the soil water content and to calculate soil bulk density. Soil water retention curves

were obtained with pressure plates and soil hydrophobicity was assessed with the Water Drop Penetration Test (WDPT); ecophysiological measurements of leaf water potentials and leaf gas exchange were made in each experimental plot in summer 2011 on two cloud-free days. The methodology and protocols for these measurements are extensively described in Baronti et al. (2014).

Grapes were sampled in 2009 (only B and C plots), and in 2010, 2012 and 2013 (B, BB and C plots). For quantitative analysis 5 plants for each plot were hand-harvested (a total of 30 sampled plants in 2009 and 45 sampled plants in 2010, 2012, 2013), clusters per plant were counted and weighted individually, a sample of 50 berries for each plot was weighted and used to calculate skin-to-pulp ratio and number and weight of seeds (only in 2012 and 2013).

From the harvested plants a sample of 2.5 kg of grapes was taken for the analysis of following quality parameters: soluble solids (expressed as °Brix), Total Acidity (expressed as g $l^{-1}$of tartaric acid) pH, and Total Anthocyanins (mg l^{-1}). Sampled clusters were crunched with a laboratory press and musts analysed with a Foss analyser (WineScan^{TM}, Foss, Denmark).

Statistical analysis was performed in R statistical environment (R development team, 2014). Post-hoc comparison adopting one-way ANOVA scheme were made with "multicomp" R package (Hothorn et al., 2008). Tukey HSD testing framework was used to verify the significance of differences between treatments (C, B, BB).

3. Results

The addition of biochar to soil caused a substantial and significant change in soil physical characteristics with a decrease of soil bulk density and an increases in available soil water content in treated soils compared to control soils (from 3.2% to 45% in the 22 and 22+22 t ha^{-1} application

140 rates, respectively. These changes translated into increased leaf water potential (24–37%) during
141 droughts (Baronti et al., 2014).
142 Results of fruit sampling made at harvest indicate that grape yield per plant significantly increased
143 in biochar treated plots in all harvest years (Table 3). The proportional increase relative to the
144 control ranged from 16% to 66% lasting across the four years of the experiment; the biochar re-
145 application treatment did not result in any significant improvement.
146 The number of clusters per plant was not affected, while the average cluster weights were always
147 significantly increased, except in the first year, with a maximum of 46% in BB plots in 2012 (Table
148 4); no significant differences in average cluster weight were observed between B and BB
149 treatments. Plants from treated plots showed significantly bigger berries in 2012 (B) and in 2013
150 (both B and BB), with a maximum increase of 14.8% in BB plot in 2013, compared to controls
151 (Table 5).
152 Analysing the yield difference in the four harvests between the control and the biochar treatments
153 (pooled together), the observed increase in productivity was inversely correlated (R^2=0.75) with the
154 precipitation recorded during the growing season (Figure 2). The effect of biochar on yield was
155 therefore higher in the years with lowest rainfall.
156 Grape quality parameters show a high inter-annual variability confirming the importance of climatic
157 drivers on grapevine quality. The observed increase of productivity did not correspond to a
158 significant variation in any of the selected quality parameters between treatments and their controls
159 (Table 6).
160 The analysis of seeds made in 2012 and 2013, highlighted a significant increase in number of seeds
161 per berry (up to 29.8% for BB in 2013) and total seed weight per berry (up to 28.7% for B in 2012)
162 in treated plots respect to the control, with no changes in average seed weight (Table 7).
163
164 **4. Discussion**

Water availability plays a key role in determining the quality and quantity of most agricultural production in the Mediterranean region and based on our results this includes vineyard production. Water deficits are known to alter fruit mass and composition mainly through direct effects on the berry size; in general higher water availability translates into increased production with detrimental effect on key grape quality parameters (Van Leeuwen et al., 2009; Bramley et al., 2011). With full water availability, photosynthetic activity is not constrained by stomata closure and grape sugar levels are lower due to competition for carbon substances between berry and shoots and because of the greater sugar dilution in a bigger berry associated to increased water content; on the contrary, in mild water deficit conditions ripening is promoted by the production of abscisic acid in roots and vine produces smaller berries higher in sugar, anthocyanins and tannins content (Van Leeuwen et al., 2009).

In our experiment, biochar application increased soil water content and plant available water, this is likely to have driven the substantial increase in productivity (yield, average cluster weight and berry size) in all harvests. This effect was higher in the years with lowest rainfall, convincingly supporting the idea of a positive regulation effect of biochar in case of water shortages. Unexpectedly, no significant effects were observed on key grape quality parameters, this suggests that the increased plant water availability due to biochar has a complex mechanism of action on plant physiology and involves an effect on tissues formation. In fact, it is recognized that the consequences of changes of berry size on overall fruit quality are not linear and that the timing of water stress affects the mass ratio of berry seeds, skin and pulp (Roby and Mattews, 2004); early water deficits (before *veraison*) may have an effect on the final cell number by reduced growth of mesocarp tissues (Considine and Knox, 1981); later on, during ripening, water stress can affect cell extension and overall berry hydration. The observed significant increase in seed mass, in response to an increased number of seeds per berry (from 1.6 to 2.02 in 2012 and from 1.45 to 1.88 in 2013 for C and BB treatments, respectively), confirms the key role of prior-to-veraison water stress in the fruit set and tissue formation (Roby and Mattews, 2004; Korkutal et al., 2011).

191 Grape fertility is also driven by other factors that could have been influenced by biochar, such as
192 the nitrogen availability in the preceding season (Duchene et al., 2011), when it is well known that
193 biochar affects nitrogen cycling and reduces its leaching (Ventura et al., 2013), and by temperature
194 regimes near flowering (Ebadi et al., 1996) when it was proved that the changes of soil reflectivity
195 (albedo) that follows biochar application may affect energy fluxes partitioning with a positive
196 increase of soil temperatures (Genesio et al., 2012) and a consequent enhanced temperature regime
197 at canopy level.
198 Biochar application is therefore likely to have driven the final fruit composition and mass through
199 multiple mechanisms with different timing of action, namely i) a reduction in nitrogen leaching in
200 the preceding season that promoted fruit set and seeds number, ii) an increase of temperature
201 regimes near flowering that favoured grape fertility iii) a reduction of water stress in pre-veraison
202 that favoured multiplication of tissues cells and therefore the thickness of skin and iv) a reduction in
203 evaporative loss of berry in the latest phases of ripening due to the higher plant water availability.
204 The ensemble of those mechanisms could explain the unexpected results in grape quality
205 parameters where, despite the diminished water availability of non-treated vines, the berries of the
206 control vines did not show neither significantly higher sugars nor an increase in anthocyanin
207 concentration. Indeed, the observed differences in yield did not correlate with changes in grape
208 quality parameters; coherently, similar results were found in the skin-to-pulp ratio where no-
209 significant differences between treatments were observed (data not shown).
210 The lack of significant effect on grape quality parameters is in agreement with the findings of
211 Schmidt et al. (2014) that did not observe any significant difference between treatments and
212 controls after the application of 8 t ha^{-1} of biochar to a vineyard in Valais, although the differences
213 in soil and biochar application rate and the absence of water limiting conditions in the Valais
214 vineyard limits the possibility to compare the two experiments.
215 The absence of significant yield differences between the two biochar treatments suggests that a
216 saturation response was already reached at the application of 22 t ha^{-1}, in agreement with the

observed saturation on plant water relationships (Baronti et al., 2014). In the specific conditions of our experiment, the biochar saturation threshold is likely to be lower than 22 t ha^{-1}. Moreover, in our experiment, biochar benefits did not have a transient nature as observed elsewhere after biochar reapplication (Quillian et al., 2012) but, on the contrary, its effect was maintained for at least four years.

These findings are particularly relevant when seen in the context of the expected changes in climate and the required adaptation strategies that should be urgently considered for a sustainable viticulture (Webb et al., 2012; Diffenbaugh et al., 2011). Indeed, the quality and quantity of grapevine production remains strictly linked to seasonal climate variability. While it has been proven that moderate water stress might lead to improved grape quality, excessive stress conditions are known to lead to an imbalance of the sugar/acidity ratio and to an impairment of the dynamic of polyphenols accumulation, causing major difficulties in the identification of the optimal harvest time (Jones et al., 2005). For this reason irrigation has been considered a key strategy to reduce the impact of climate variability and to enhance quality also through regulated deficit irrigation and partial root drying approaches (Bindon et al., 2008; Romero and Martinez-Cutilla, 2012). But irrigation inevitably competes with other civil and industrial water uses and water availability may become scarce especially if the expected reduction in precipitation in some important viticulture areas of the world will be confirmed (Hannah et al., 2013). In addition, irrigation is known to have negative impact on soil erosion and groundwater quality and might remain banned in many AOC areas (Appellation d'Origine Controllée) or limited to extreme emergency situations. The use of biochar as a soil amendment could therefore become an effective adaptation strategy and alternative to irrigation, to enhance the drought tolerance of vineyards leading to substantial increases in yields without any evident detrimental effect on quality parameters.

The definition of such an adaptive strategy certainly calls for further studies to provide more focused insights in molecular and physiological mechanisms associated to biochar application to

vineyards and to better account for the biochar specificity of action in different areas regions and soils.

In particular the increased grape fertility, expressed by the number of seeds per berry, observed in our experiment, must be investigated further to confirm the existence of mechanisms other than water and nutrient availability with particular reference to microbial activity and hormonal pathways (Gomez et al., 2014; Spokas et al., 2010). Moreover, changes in the susceptibility of biochar-treated vineyards to pests and diseases should also be investigated in detail, as recent transcriptional studies have shown that biochar-driven growth stimulation in model plants is associated to changes in defence genes expression (Viger et al., 2014).

5. Conclusions

The likely consequences of using biochar as a soil amendant to reduce water stress in viticulture do require some careful analysis and detailed impact assessment. Our study dealt specifically with the assessment of the impact of high rates of biochar application on vine yield and grape quality parameters. The main conclusions are that:

i) biochar substantially increased vineyard production in all harvest-years; this yield increase was inversely correlated with rainfall in the vegetative period, thus highlighting that the effect of biochar was higher in dry years and emphasising the role of biochar in increasing plant water availability;

ii) despite the yield increase, no detrimental effects on key grape quality parameters were observed; the absence of qualitative differences suggests that biochar mechanisms of action is more complex than being the direct consequence of the improved water status and, although some hypothesis can be formulated, the full understanding of such mechanisms requires further investigations;

iii) a biochar based strategy could be effectively adopted in vineyards in drought prone areas as an alternative to irrigation.

Finally, the profiling of wine quality attributes following biochar application need to be further explored in long term experiments by means of sensory panels and testing this strategy in other soils and wine regions.

Acknowledgements

The authors acknowledge the “Marchesi Antinori - La Braccesca Estate” for hosting the experiment and in particular Christian Zulian, Samuele Collini and Giancarlo Certini for their technical support in field campaigns and Alfonso Crisci (IBIMET CNR) for the support in defining the statistical approach. This work contributes to the Eurochar project (FP7-ENV-2010 ID-265179) and AgroPyroGas project (Regione Toscana POR CRO FSE 2007–2013 asse IV). The Italian Biochar Association (ICHAR www.ichar.org) is also acknowledged.

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382

383 Table 1 - Soil properties of the experimental vineyard.

	Unit	Value
Sand [a]	g kg^{-1}	450
Silt	g kg^{-1}	200
Clay	g kg^{-1}	350
Bulk density	t m^{-3}	1.45
OC [b]	g kg^{-1}	4.7
N [c]	g kg^{-1}	0.46
K available	mg K_2O kg^{-1}	192
Ca available	mg CaO kg^{-1}	1454
Mg available	mg MgO kg^{-1}	939
Na	mg Na kg^{-1}	97
CEC [d]	meq $100g^{-1}$	25.7
pH [e]		5.37

384 a) refers to fine (<2 mm) texture fraction.
385 b) Organic Carbon (OC) content was determined using a CHN auto-analyser (CHN 1500, Carlo Erba).
386 c) Nitrogen (N) content was determined using a CHN auto-analyser (CHN 1500, Carlo Erba).
387 d) Cation Exchange Capacity (CEC) was determined using the NH4OAc method
388 e) pH was measured in a 1:2.5 (mass/vol) soil solution.
389

390 Table 2 - Chemical and physical characteristics of biochar applied in the field experiment (modified
391 from Baronti et al., 2014).

	Unit	Value
C	%	77.81
N	%	0.91
Al	mg kg^{-1}	268
C/N	-	63.53
Ca	mg kg^{-1}	25000
Cu	mg kg^{-1}	97
Fe	mg kg^{-1}	333
K	mg kg^{-1}	13900
Mg	mg kg^{-1}	28700
Mn	mg kg^{-1}	84
Na	mg kg^{-1}	11900
P	mg kg^{-1}	23300
S	mg kg^{-1}	481
Zn	mg kg^{-1}	104
pH	-	9,8
CEC	(cmolc/kg).	101
Max water absorption	g g^{-1} of d. m.	4.53
BET	m^2 g^{-1}	410 ± 6
Total porosity	mm^3 g^{-1}	2722
Transmission pores	mm^3 g^{-1}	318
Storage pores	mm^3 g^{-1}	1997
Residual pores	mm^3 g^{-1}	406
Particle size distribution (mm):		
50-20	%	4.45
20-10	%	12.1
10-8	%	13.1
8-4	%	10.36
4-2	%	19.85
2-1	%	24.2
<1	%	15.94

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394 Table 3 - Grape yield per plant (fresh weight). Δy is the % yield variation in biochar treated plots
395 with respect to the control (s.e. is the standard error). All the values are average of 25 plants for
396 each treatment per year. Values followed by the same letters are not statistically different at P =
397 0.05 by the Student–Newman–Keuls test.

Year	Treat.	Yield ± s.e (kg plant^{-1})	Sign. code	Δy (%)
2009	C	1.36 ± 0.08	a	
	B	1.63 ± 0.09	b	20
2010	C	1.34 ± 0.09	a	
	B	2.12 ± 0.19	b	58.1
	BB	1.90 ± 0.16	b	42.3
2012	C	1.05 ± 0.09	a	
	B	1.62 ± 0.14	b	54.6
	BB	1.75 ± 0.14	b	66.8
2013	C	1.44 ± 0.11	a	
	B	1.68 ± 0.11	ab	16.1
	BB	1.95 ± 0.15	b	35.3

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400 Table 4 - Number of clusters per plant and average cluster fresh weight at harvest. Δy is % of
401 variation in biochar treated plots respect to the control (s.e. is the standard error). Values are
402 average of 25 plants for each treatment. Values followed by the same letters are not statistically
403 different at P = 0.05 by the Student–Newman–Keuls test.

Year	Treat.	N° clusters plant^{-1} ± s.e.	Sign. codes	Δy (%)	Avg. cluster weight ± s.e (g)	Sign. codes	Δy (%)
2009	C	14.47 ± 0.40	a		106.44 ± 5.99	a	
	B	14.33 ± 0.51	a	-1.0	115.94 ± 5.46	a	8.9
2010	C	15.64 ± 0.73	a		85.56 ± 2.47	a	
	B	18.12 ± 0.97	a	15.9	116.49 ± 3.39	b	36.2
	BB	17.90 ± 1.04	a	14.5	106.37 ± 3.53	b	24.3
2012	C	14.80 ± 0.83	a		71.02 ± 2.25	a	
	B	16.22 ± 0.92	a	9.6	100.33 ± 3.09	b	41.3
	BB	16.82 ± 1.08	a	13.6	104.35 ± 2.84	b	46.9
2013	C	18.00 ± 1.15	a		80.05 ± 2.01	a	
	B	18.12 ± 0.96	a	0.7	92.53 ± 2.42	b	15.6
	BB	18.76 ± 1.44	a	4.2	99.72 ± 2.53	b	24.7

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406 Table 5 - Fresh weight of 50 berries at harvest. Δy is the %variation in biochar treated plots with
407 respect to the control (s.e. is the standard error). Values followed by the same letters are not
408 statistically different at P = 0.05 by the Student–Newman–Keuls test.

Year	Treat.	Fresh weight 50 berries ± s.e (g)	Sign. codes	Δy (%)
2010	C	63.20 ± 5.28	a	
	B	66.21 ± 5.79	a	4.8
	BB	67.02 ± 5.93	a	6.0
2012	C	46.92 ± 1.12	a	
	B	51.52 ± 1.46	b	9.8
	BB	49.40 ± 1.12	a	5.3
2013	C	81.20 ± 2.52	a	
	B	88.40 ± 2.50	b	8.9
	BB	93.20 ± 3.14	b	14.8

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411 Table 6 - Grape quality parameters (°Brix; Total Acidity=AT; pH; °Brix/AT; Total Anthocyanins=

412 ANT) at harvest for C, B and BB treatments in 2009, 2010, 2012 and 2013. Values followed by the

413 same letters are not statistically different at P = 0.05 by the Student–Newman–Keuls test.

Year	Treat.	°Brix	Sign. codes	AT	Sign. codes	pH	Sign. codes	°Brix/AT	Sign. codes	ANT	Sign. codes
2009	C	24.65±0.33	a	5.54±0.19	a	3.42±0.01	a	4.47±0.14	a	1123±44	a
	B	23.95±0.53	a	5.56±0.18	a	3.42±0.02	a	4.33±0.19	a	1186±73	a.
2010	C	24.74±0.2	a	6.82±0.19	a	3.32±0.01	a	3.64±0.08	a	1024±24	a
	B	25.02±0.24	a.	7.3.2±0.47	a.	3.36±0.01	a	3.49±0.25	a	1045±41	a
	BB	24.82±0.22	a	6.64±0.34	a	3.34±0.01	a	3.78±0.22	a	1019±29	a
2012	C	24.32±0.12	a	4.42±0.05	a	3.65±0.01	a	5.50±0.06	a	937±48	a
	B	24.08±0.26	a	4.22±0.09	a	3.62±0.02	a	5.71±0.12	a	949±75	a
	BB	23.92±0.26	a.	4.1±0.13	a	3.62±0.02	a	5.86±0.18	a	994±.51	a
2013	C	23.51±0.07	a	5.83±0.14	a	3.36±0.01	a	4.04±0.11	a	1143±38	
	B	23.23±0.17	a.	6.08±0.12	a.	3.36±0.01	a	3.83±0.09	a	1038±68	a
	BB	23.19±0.28	a	5.87±0.14	a	3.39±0.01	a	3.96±0.13	a	1001±70	a

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Table 7 – Seeds number for 50 berries (SN), seeds dry weight for 50 berries (SDW) and average seed weight (ASW) for the Control (C) and treated plots (B and BB) in 2012 and 2013. Δy is the SN, SDW and ASW variation (%) in biochar treated plots respect to the control (s.e. is the standard error). Values followed by the same letters are not statistically different at P = 0.05 by the Student–Newman–Keuls test.

Year	Treat.	SN ± s.e.	Sign codes	Δy (%)	SDW ± s.e. (g)	Sign codes	Δy (%)	ASW ± s.e. (g)	Sign. codes	Δy (%)
2012	C	82.20 ± 6.37	a		2.03 ± 0.13	a		$2.48\ 10^{-2} \pm 4.19\ 10^{-4}$	a	
	B	106.10 ± 5.64	b	29.0	2.62 ± 0.14	b	28.7	$2.46\ 10^{-2} \pm 1.58\ 10^{-4}$	a	-0.7
	BB	102.90 ± 5.13	ab.	25.2	2.45 ± 0.14	ab	20.7	$2.38\ 10^{-2} \pm 4.95\ 10^{-4}$	a	-4.2
2013	C	72.60 ± 3.33	a		2.33 ± 0.11	a		$3.21\ 10^{-2} \pm 7.49\ 10^{-4}$	a	
	B	80.20 ± 3.02	a	10.5	2.50 ± 0.08	a.	7.0	$3.11\ 10^{-2} \pm 4.93\ 10^{-4}$	a	-3.1
	BB	94.20 ± 2.15	b	29.8	2.89 ± 0.10	b	23.8	$3.06\ 10^{-2} \pm 5.49\ 10^{-4}$	a	-4.8

Figure 1

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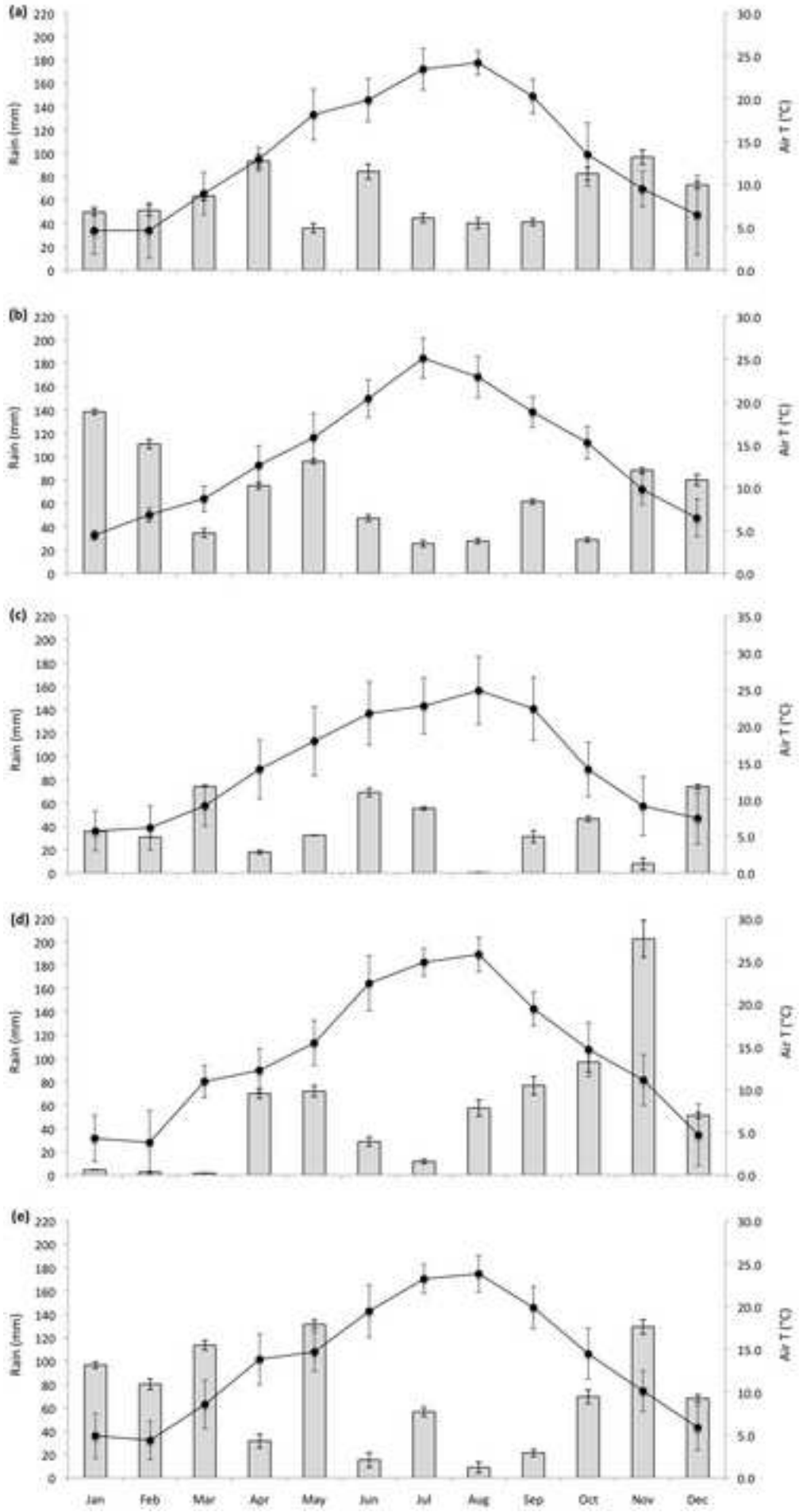


Figure 2

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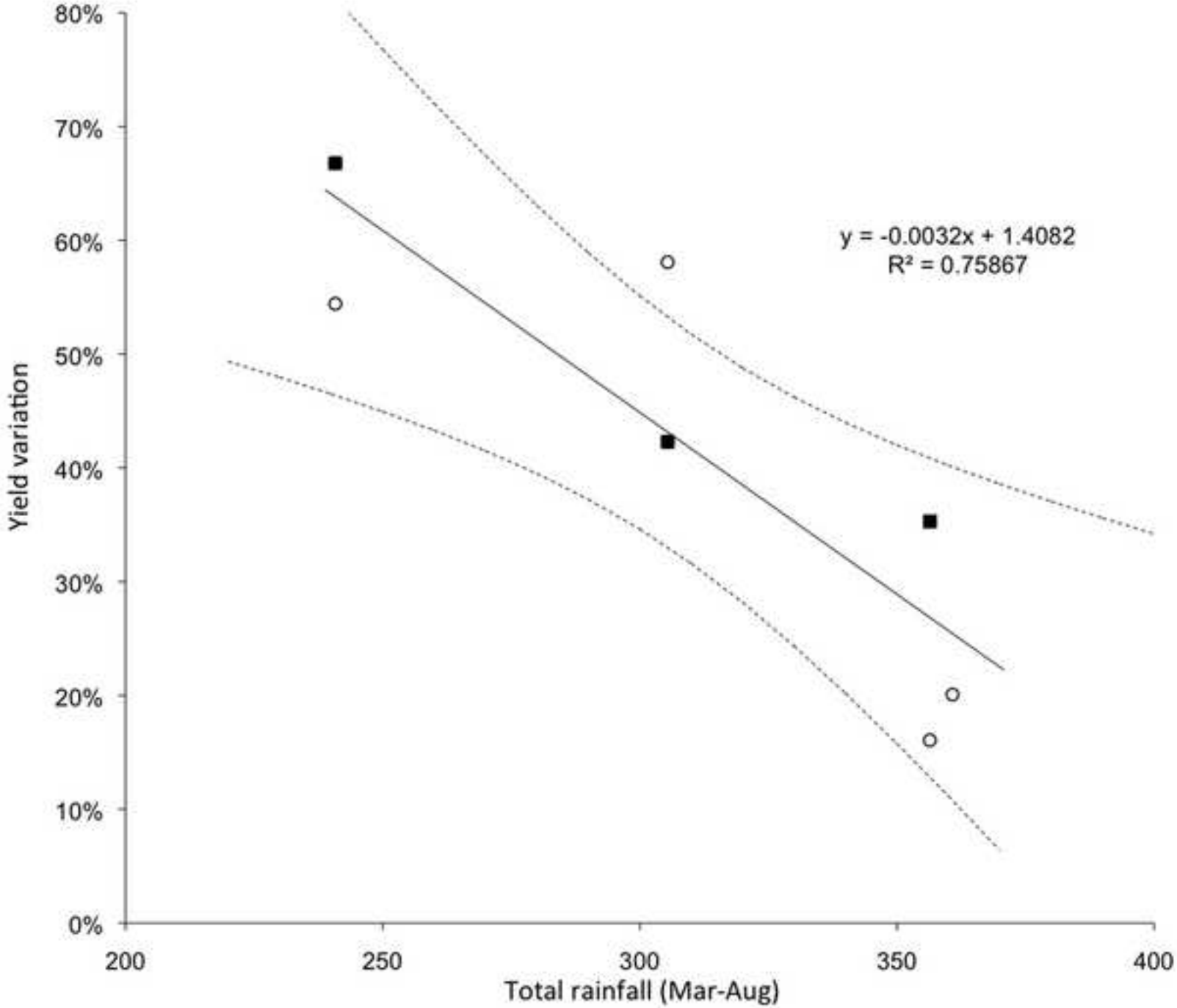


Figure
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Figure 1: summary of monthly cumulated rainfall (colums, left axis)) and monthly average temperature (dots, right axis) for the years 2009 (a), 2010 (b), 2011 (c), 2012 (d), 2013 (e). Error bars represent standard error.

Figure 2: regression between total rainfall in the vegetative period (March-August) and the % yield variation of treated plots versus control (BB-C= black blocks; B-C=white circles). Trend line (solid, black) pools together BB and B data. Dashed lines represents the 95% confidence interval.